quently, perfect discrimination is impossible for non-orthogonal states. Insights such as these are prevalent when applying state discrimination to problems in quantum information theory. The fact that we cannot perfectly distinguish between non-orthogonal states is what makes quantum key distribution possible.

Another important use of state discrimination is to characterize properties of quantum states, such as entanglement and separability. Much research has been devoted to this area. One of the main motivations behind such studies is to determine whether entangled states are useful for certain tasks, or whether entanglement is needed for a given task at all. Quantum metrology, quantum imaging, and the classical capacity of quantum channels are examples of such investigations.

State discrimination is also indispensable in quantum error correction. Error correction cannot be performed without syndrome measurement. A syndrome measurement determines whether a quantum state has been corrupted and, if so, which part of the state has been affected. The syndrome measurement is typically a multi-state measurement that does not disturb the encoded state but still provides information about the error. Such measurements lie at the heart of error correction.

Finally, state discrimination has found uses in quantum algorithms through a variant known as quantum state filtering. In quantum state filtering, the goal is to determine whether an unknown quantum state is a specific state $|\psi_k\rangle$ from a set of known states $\{|\psi_1\rangle, \ldots, |\psi_N\rangle\}$, or one of the others. Using optimal solutions in quantum state filtering, it is possible to implement an efficient probabilistic quantum algorithm that distinguishes between sets of Boolean functions, which generalizes the Deutsch-Jozsa algorithm. It has also been proposed that quantum decision problems can be characterized by state discrimination.

Having discussed the role of state discrimination in the major topics of quantum information theory, we now introduce the encoding of information onto continuous variable states.

1.5 Continuous Variables in Quantum Mechanics

While quantum information is primarily concerned with discrete quantum variables, such as qubits, continuous variable states have always played an important role in quantum mechanics. The position and momentum of the quantized harmonic oscillator are continuous variables. Because a mode of the electromagnetic field can be thought of as a harmonic oscillator, its quadrature amplitudes (the electric and magnetic fields) are continuous

variables. Due to this connection to the quantized electromagnetic field, the development of the laser in the 1960s inspired considerable research into the properties of continuous quantum variables.

Continuous variables also play an important role in quantum information. The original EPR thought experiment was formulated in terms of the entangled momentum and position states of two particles. Furthermore, one of the first experimental demonstrations of quantum teleportation was performed using continuous variable states. Many aspects of discrete quantum information theory have continuous analogues, including quantum codes. It is even possible to formulate a theory of universal quantum computation over continuous variables by combining linear optical devices such as phase shifters and beamsplitters with nonlinear elements such as downconverters and parametric amplifiers.

The primary reason for developing a theory of quantum information over continuous variables is a practical one. Most communication channels involve the exchange of bosons (such as photons or phonons), which correspond to quantized harmonic oscillators. Accordingly, the theory of quantum information over continuous variables is essential for understanding the physical limits of communication. Moreover, many important sensing and measurement techniques, including microscopy and radar, rely on bosonic systems. Quantum information over continuous variables is the key to understanding the limits of sensing and measurement and for devising methods to attain those limits. With this motivation, this work explores the fundamental limits of quantum communication and sensing and introduces techniques to achieve them, beginning with a review of continuous variable quantum information theory in the following chapter.

1.6 Scope of This Work

This subsection lists the major research areas in continuous variable quantum computing and quantum information, and situate the contributions of this work within them.

Major areas of research include:

1. **Channel Capacity of the Noisy, Lossy Bosonic Channel:** A fundamental question in quantum communication is: what is the maximum rate at which information can be sent down a bosonic channel in the presence of loss and noise? The capacity of a quantum channel $\mathcal{E}$ with inputs drawn from an ensemble of states $\hat{\rho}_k$ with probabilities p_k is given by its Holevo quantity,

$$\chi(\{p_k, \hat{\rho}_k\}) = S\left(\sum_k p_k \mathcal{E}(\hat{\rho}_k)\right) - \sum_k p_k S\left(\mathcal{E}(\hat{\rho}_k)\right), \tag{1.37}$$

where $S(\hat{\rho})$ is the von Neumann entropy of the state $\hat{\rho}$. Maximizing this quantity over all possible ensembles, including entangled inputs across multiple uses of the channel, yields the channel capacity. This maximization is often difficult to perform. The capacity of the bosonic channel with linear loss is known and is attained by sending ensembles of coherent states down the channel. By contrast, the capacity of the bosonic channel with linear loss and thermal noise remains unknown, although it is conjectured to also be attained by sending ensembles of coherent states.

2. **Minimum Output Entropy Conjecture:** The capacity of the bosonic channel with linear loss and thermal noise is indeed attained by coherent state inputs if the bosonic minimum output entropy conjecture holds. This conjecture states that a coherent-state input yields the output with the minimum von Neumann entropy for a bosonic channel with additive Gaussian noise. Proving or disproving the bosonic minimum output entropy conjecture is perhaps the most important open problem in continuous variable quantum information theory.

3. **Optimal Measurements for Distinguishing Bosonic States:** Distinguishing quantum states is an important component of quantum information science. Key results, such as the no-cloning theorem, the quantum teleportation protocol, and more recently the Yuen no-go theorem, have their origins in problems of state discrimination. The Helstrom bound gives the minimum probability of error technique for discriminating between two quantum states. The quantum Chernoff bound is an upper bound to the Helstrom limit of multiple-copy binary state discrimination, which is asymptotically tight in the error exponent. While the measurement techniques for attaining the Helstrom bound are known in principle, devising experimentally practical measurement techniques for attaining the Helstrom bound is not known in general. Moreover, even when the channel capacity for a quantum channel is attained by product state (unentangled) inputs, the measurement required to attain the capacity is the entangling 'pretty-good' measurement, which may be hard to perform in practice. For example, even though the channel capacity for the bosonic channel with linear loss is attained by coherent state inputs, practical decoding techniques that attain this bound are not currently known.

2 Continuous Variable Quantum Information Theory

2.1 Introduction

In Continuous Variable (CV) Quantum Information Theory, one goes beyond discrete systems (such as qubits) to systems described by **continuous degrees of freedom** . Many physical systems are naturally continuous. Examples include: Position and momentum of a particle, electromagnetic field modes, vibrational modes of trapped ions. We traditionally formulate quantum information theory using qubits. Therefore, concepts in CV quantum information are often less familiar and requires - building a unified formalism, which is the main purpose of this review.
As we mentioned, many realistic quantum systems are continuous. A mode of the electromagnetic field behaves like a **quantum harmonic oscillator**. Studying the quantization of electromagnetic fields is important, as optical systems are widely used in communication and sensing, considering the experimental accessibility of photons. The theoretical framework we would describe here would be based on: the quantum harmonic oscillator, Creation and annihilation operators, various quantum gates and phase space representations (e.g., Wigner function). The tools built would allow us to describe quantum states in phase space, analyse Gaussian states, and study transformations via linear operations. Ultimately, the formalism developed here would be used for explaining bosonic quantum channels and state discrimination later.

2.2 Quantization of a Single Electromagnetic Field Mode

In this section, we will derive the Hamiltonian corresponding to a single mode of the electromagnetic field confined in a cavity. We will show that each mode behaves as a simple harmonic oscillator.

Consider a single mode of the electromagnetic field in a cavity of volume $V = AL$, where A is the cross-sectional area and L is the length along the z-direction.

We assume: The field propagates along the z-axis, the electric field is polarized along the x-direction, the magnetic field is along the y-direction.
The electric field is given by:

$$\mathbf{E}(z,t) = \hat{x}\sqrt{\frac{\omega^2}{4\epsilon_0 V}}\, q(t)\sin(kz) \tag{2.1}$$

The magnetic field is:

$$\mathbf{B}(z,t) = \hat{y}\sqrt{\frac{1}{4\epsilon_0 V}}\, \dot{q}(t)\cos(kz) \tag{2.2}$$

where:

- $q(t)$ is the generalized coordinate (field amplitude)
- $\dot{q}(t) = \frac{dq}{dt}$
- $\omega = ck$ is the angular frequency
- ϵ_0 is the permittivity of free space

2.2.1 Hamiltonian of the Field

The total energy of the electromagnetic field is:

$$H = \frac{1}{2}\int_V dV \left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) \tag{2.3}$$

Substituting **E**:

$$E^2 = \frac{\omega^2}{4\epsilon_0 V} q^2(t)\sin^2(kz)$$

$$\epsilon_0 E^2 = \frac{\omega^2}{4V} q^2(t)\sin^2(kz)$$

Now for **B**:

$$B^2 = \frac{1}{4\epsilon_0 V}\dot{q}^2(t)\cos^2(kz)$$

Using $\frac{1}{\mu_0\epsilon_0} = c^2$:

$$\frac{1}{\mu_0}B^2 = \frac{c^2}{4V}\dot{q}^2(t)\cos^2(kz)$$

Since $\omega = ck$:

$$\frac{1}{\mu_0}B^2 = \frac{\omega^2}{4Vk^2}\dot{q}^2(t)\cos^2(kz)$$

Thus, we have the Total Hamiltonian in eq. (2.3) as:

$$H = \frac{1}{2}\int_V dV \left[\frac{\omega^2}{4V}q^2(t)\sin^2(kz) + \frac{c^2}{4V}\dot{q}^2(t)\cos^2(kz)\right]$$

Using:

$$\int_0^L \sin^2(kz)\,dz = \frac{L}{2}, \quad \int_0^L \cos^2(kz)\,dz = \frac{L}{2}$$

we obtain:

$$\int_V \sin^2(kz)\,dV = \frac{V}{2}, \quad \int_V \cos^2(kz)\,dV = \frac{V}{2}$$

Substituting:

$$H = \frac{1}{2}\left[\frac{\omega^2}{4V}q^2(t)\cdot\frac{V}{2} + \frac{c^2}{4V}\dot{q}^2(t)\cdot\frac{V}{2}\right]$$

Simplifying:

$$H = \frac{1}{2}\left[\frac{\omega^2}{8}q^2(t) + \frac{c^2}{8}\dot{q}^2(t)\right]$$

we rewrite:

$$H = \frac{1}{16}\left[\omega^2 q^2(t) + c^2\dot{q}^2(t)\right] \tag{2.4}$$

We now absorb the constants into the variables by redefining:

$$q(t) \to \frac{\omega}{4\sqrt{\hbar\omega}}\,q, \qquad \dot{q}(t) \to \frac{c}{4\sqrt{\hbar\omega}}\,\dot{q}(t)$$

Substituting in eq. (2.4), we obtain:

$$H = \hbar\omega(q^2 + \dot{q}^2) = \hbar\omega(q^2 + p^2) \tag{2.5}$$

where $p = \dot{q}$

Thus, a single mode of the electromagnetic field behaves as a simple harmonic oscillator with frequency ω. This forms the basis for quantization.

2.3 Bosonic Modes from the Harmonic Oscillator

From the previous section, the Hamiltonian of a single electromagnetic field mode can be written as:

$$\hat{H} = \hbar\omega(\hat{q}^2 + \hat{p}^2)$$

where $\hat{q}$ and $\hat{p}$ are dimensionless quadrature operators.

2.3.1 Connection to the Standard Harmonic Oscillator

We now relate the dimensionless operators $(\hat{q}, \hat{p})$ to the physical position and momentum operators $(\hat{x}, \hat{p}_x)$ of the harmonic oscillator of mass m, with oscillation frequency ω.

Define:

$$\hat{q} = \sqrt{\frac{m\omega}{2\hbar}}\,\hat{x}, \qquad \hat{p} = \frac{1}{\sqrt{2m\hbar\omega}}\,\hat{p}_x \tag{2.6}$$

Substituting into the Hamiltonian in eq. (2.5):

$$\begin{aligned}\hat{H} &= \hbar\omega\left(\frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\hbar\omega}\hat{p}_x^2\right)\\ &= \frac{1}{2}m\omega^2\hat{x}^2 + \frac{\hat{p}_x^2}{2m}\end{aligned}$$

Thus, we recover the standard harmonic oscillator Hamiltonian:

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

In quantum mechanics, the position and momentum operators satisfy:

$$[\hat{x}, \hat{p}_x] = i\hbar$$

Using the definitions of $\hat{q}$ and $\hat{p}$, we compute:

$$\begin{aligned}
[\hat{q}, \hat{p}] &= \left[\sqrt{\frac{m\omega}{2\hbar}}\hat{x}, \frac{1}{\sqrt{2m\hbar\omega}}\hat{p}_x\right] \\
&= \frac{1}{2\hbar}[\hat{x}, \hat{p}_x] \\
&= \frac{1}{2\hbar}(i\hbar) \\
&= \frac{i}{2}
\end{aligned}$$

Thus, the quadrature operators satisfy:

$$[\hat{q}, \hat{p}] = \frac{i}{2} \tag{2.7}$$

2.3.2 Creation and Annihilation Operators

We now define the bosonic ladder operators:

$$\hat{a} = \hat{q} + i\hat{p}, \qquad \hat{a}^\dagger = \hat{q} - i\hat{p}$$

They satisfy the commutation relation:

$$\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= [\hat{q} + i\hat{p}, \hat{q} - i\hat{p}] \\
&= -i[\hat{q}, \hat{p}] + i[\hat{p}, \hat{q}] \\
&= -i\left(\frac{i}{2}\right) + i\left(-\frac{i}{2}\right) \\
&= 1
\end{aligned}$$

2.3.3 Hamiltonian in Terms of Ladder Operators

Using:

$$\begin{aligned}\hat{a}^\dagger \hat{a} &= (\hat{q} - i\hat{p})(\hat{q} + i\hat{p}) \\ &= \hat{q}^2 + i\hat{q}\hat{p} - i\hat{p}\hat{q} - i^2\hat{p}^2 \\ &= \hat{q}^2 + \hat{p}^2 + i(\hat{q}\hat{p} - \hat{p}\hat{q}) \\ &= \hat{q}^2 + \hat{p}^2 + i[\hat{q}, \hat{p}] \\ &= \hat{q}^2 + \hat{p}^2 - \frac{1}{2}\end{aligned}$$

we obtain:

$$\begin{aligned}\hat{H} &= \hbar\omega(\hat{q}^2 + \hat{p}^2) \\ &= \hbar\omega\left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right)\end{aligned}$$

Physical Interpretation: The operators $\hat{a}$ and $\hat{a}^\dagger$ describe annihilation and creation of quanta of the electromagnetic field. Each mode behaves as a quantum harmonic oscillator, and its energy eigenstates correspond to photon number states.

2.3.4 Fock States and Vacuum State

We define the vacuum state $|0\rangle$ as:

$$\hat{a}\,|0\rangle = 0 \tag{2.8}$$

Higher number states are generated by repeated application of the creation operator:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle \tag{2.9}$$

These states satisfy:

$$\hat{N}\,|n\rangle = n\,|n\rangle \tag{2.10}$$

where we have defined the number operator as: $\hat{N} = \hat{a}\hat{a}^\dagger$ The annihilation and creation operators act as:

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \quad n \geq 1$$

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

Thus, the Fock states represent states with a definite number of quanta (photons).

2.3.5 Multi-Mode Bosonic System

We now generalize to a system of n independent modes.

Each mode is described by its own annihilation operator:

$$\hat{a}_k, \quad k = 1, 2, \ldots, n$$

These operators satisfy the canonical commutation relations:

$$[\hat{a}_k, \hat{a}_l^\dagger] = \delta_{kl} \tag{2.11}$$

The Hilbert space of the system is the tensor product of the individual Fock spaces' hamiltonians:

$$\mathcal{H} = \bigotimes_{k=1}^{n} \mathcal{H}_k \tag{2.12}$$

The number operator for each mode is:

$$\hat{N}_k = \hat{a}_k^\dagger \hat{a}_k \tag{2.13}$$

The total Hamiltonian of the system is:

$$\hat{H} = \sum_{k=1}^{n} \hbar\omega_k \left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2} \right) \tag{2.14}$$

2.3.6 Quadrature Operators for Each Mode

For each mode, we define the quadrature operators:

$$\hat{q}_k = \frac{1}{2}(\hat{a}_k + \hat{a}_k^\dagger)$$

$$\hat{p}_k = \frac{1}{2i}(\hat{a}_k - \hat{a}_k^\dagger)$$

These satisfy the commutation relation:

$$[\hat{q}_k, \hat{p}_l] = \frac{i}{2}\delta_{kl}$$

2.3.7 Phase Space Representation

We now introduce the vector of quadrature operators:

$$\hat{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2, \ldots, \hat{q}_n, \hat{p}_n)^T$$

The commutation relations can be compactly written as:

$$[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij} \tag{2.15}$$

where Ω is the symplectic form:

$$\Omega = \bigoplus_{k=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{2.16}$$

The symplectic form Ω can be written explicitly as a $2n \times 2n$ matrix:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & 0 & \cdots & -1 \end{pmatrix} \tag{2.17}$$

This matrix encodes the canonical commutation relations between all quadrature operators and defines the symplectic structure of the phase space.

2.4 Mean Vector, Expectation Values and the Covariance Matrix

We now complete the phase-space description by defining expectation values.

For a quantum system described by a density operator $\hat{\rho}$, the expectation value of any operator $\hat{O}$ is defined as:

$$\langle \hat{O} \rangle = \mathrm{tr}(\hat{\rho}\hat{O}) \tag{2.18}$$

Here, the density operator $\hat{\rho}$ describes the full state of the system, which may be either pure or mixed.

The mean vector (first moments) of the quadrature operators is defined as:

$$\bar{R}_k = \langle \hat{R}_k \rangle = \mathrm{tr}(\hat{\rho}\hat{R}_k) \tag{2.19}$$

Thus, $\bar{R}$ gives the average values of all quadrature operators and represents the displacement of the state in phase space.

2.4.1 Covariance Matrix

The second moments of the quadrature operators are captured by the covariance matrix σ, defined as:

$$\sigma_{kl} = \frac{1}{2}\langle \hat{R}_k \hat{R}_l + \hat{R}_l \hat{R}_k \rangle - \langle \hat{R}_k \rangle \langle \hat{R}_l \rangle \tag{2.20}$$

Using the anticommutator $\{A, B\} = AB + BA$, this can be written as:

$$\sigma_{kl} = \frac{1}{2}\langle \{\hat{R}_k, \hat{R}_l\} \rangle - \langle \hat{R}_k \rangle \langle \hat{R}_l \rangle \tag{2.21}$$

The covariance matrix describes: the uncertainties (variances) of quadrature operators and the correlations between different modes.

Gaussian States

A quantum state is said to be Gaussian if its phase-space representation (such as the Wigner function) is Gaussian.

In this case, the full quantum state is completely characterized by: the mean vector $\bar{R}$, and the covariance matrix σ.

When multiple modes are considered, the state is referred to as a joint Gaussian state, meaning that the joint phase-space distribution of all quadratures is a multivariate Gaussian.

Physical Significance

The mean vector determines the position of the state in phase space, while the covariance matrix determines its spread and correlations.

These quantities will become relevant in the study of Gaussian states, Gaussian transformations, and bosonic quantum channels.

The symplectic structure defined by Ω is further explored in the following section:

2.5 The Symplectic Structure of Phase Space and the Group $\mathrm{Sp}(2n, \mathbb{R})$

In the previous section, we introduced the vector of quadrature operators:

$$\hat{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2, \ldots, \hat{q}_n, \hat{p}_n)^T$$

These operators satisfy the canonical commutation relations:

$$[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij}$$

where Ω is the **symplectic form**, given by:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} = \bigoplus_{k=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

2.5.1 Definition of Symplectic Transformations

Let us define the set of all real $2n \times 2n$ matrices:

$$\mathcal{M}(2n) = \{S \mid S \text{ is a real } 2n \times 2n \text{ matrix}\}$$

A matrix $S \in \mathcal{M}(2n)$ is called **symplectic** if it satisfies:

$$S^T \Omega S = \Omega \tag{2.22}$$

This condition ensures that the transformation preserves the symplectic structure of phase space.

2.5.2 The Symplectic Group

The set of all matrices satisfying Eq. (2.22) forms a group under matrix multiplication, called the **real symplectic group**:

$$\mathrm{Sp}(2n, \mathbb{R}) = \{S \in \mathcal{M}(2n) \mid S^T \Omega S = \Omega\}$$

This group has the following properties:

- Closure: If $S_1, S_2 \in \mathrm{Sp}(2n, \mathbb{R})$, then $S_1 S_2 \in \mathrm{Sp}(2n, \mathbb{R})$
- Identity: The identity matrix I belongs to $\mathrm{Sp}(2n, \mathbb{R})$
- Inverse: If $S \in \mathrm{Sp}(2n, \mathbb{R})$, then $S^{-1} \in \mathrm{Sp}(2n, \mathbb{R})$
- Associativity: Follows from matrix multiplication

2.5.3 Real Canonical Linear Transformations

Consider a real canonical linear transformation (where quadratures obey the canonical commutation relation in eq. (2.15) of the quadrature operators:

$$\hat{R}' = S\hat{R} \tag{2.23}$$

We now compute the commutation relations of the transformed operators:

$$\begin{aligned}
[\hat{R}'_i, \hat{R}'_j] &= \left[\sum_k S_{ik}\hat{R}_k, \sum_l S_{jl}\hat{R}_l\right] \\
&= \sum_{k,l} S_{ik}S_{jl}[\hat{R}_k, \hat{R}_l] \\
&= \sum_{k,l} S_{ik}S_{jl}\left(\frac{i}{2}\Omega_{kl}\right) \\
&= \frac{i}{2}(S\Omega S^T)_{ij}
\end{aligned}$$

As it is a real canonical linear transformation, for the commutation relations to remain invariant, we require:

$$S\Omega S^T = \Omega$$

Taking transpose of both sides:

$$S^T\Omega S = \Omega \tag{2.24}$$

which is precisely the symplectic condition.

Thus, for every canonical linear transformation S, we have S $\in$ Sp$(2n, \mathbb{R})$. Conversely for any S $\in$ Sp$(2n, \mathbb{R})$, we can have a canonical linear transformation $\hat{R}' = S\hat{R}$. Thus, **real canonical linear transformations in 2n dimensions** and **real 2n x 2n symplectic matrices** have a one-to-one correspondence. Symplectic transformations (S $\in$ Sp$(2n, \mathbb{R})$) are important because: they preserve the **uncertainty relations** in quantum mechanics, and they will describe (as we will see later) **Gaussian unitary operations** in continuous-variable quantum systems.

2.6 Quadratic Hamiltonians and Gaussian Transformations

We study Hamiltonians that are at most quadratic in the quadrature operators $\hat{R} = (\hat{R}_1, \ldots, \hat{R}_{2n})^T$. The general form is:

$$\hat{H} = \frac{1}{2}\hat{R}^T G\hat{R} + \mathbf{b}^T\hat{R}, \tag{2.25}$$

where G is a real, symmetric $2n \times 2n$ matrix ($G = G^T$) and $\mathbf{b}$ is a real vector. The symmetry of G ensures Hermiticity of our Hamiltonian. Such quadratic Hamiltonians also generate *Gaussian transformations*, mapping Gaussian states to Gaussian states.

Heisenberg Evolution: The time evolution of the quadrature operators is governed by the Heisenberg equation. Using the canonical commutation relations $[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij}$ (eq. (2.15)), we derive the equation of motion for $\hat{R}$.

Since the quadrature operators $\hat{R}_m$ have no explicit time dependence, the equation of motion for the m-th component is:

$$\frac{d\hat{R_m}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{R_m}]. \tag{2.26}$$

First, consider the commutator of the quadratic term of the Hamiltonian $\hat{H} = \frac{1}{2}\hat{R}^T G \hat{R}$ $= \frac{1}{2}\sum_{k,l} G_{kl}\hat{R}_k\hat{R}_l$ with $\hat{R}_m$:

$$\begin{aligned}\left[\frac{1}{2}\sum_{k,l} G_{kl}\hat{R}_k\hat{R}_l, \hat{R}_m\right] &= \frac{1}{2}\sum_{k,l} G_{kl}\left(\hat{R}_k[\hat{R}_l, \hat{R}_m] + [\hat{R}_k, \hat{R}_m]\hat{R}_l\right) \\ &= \frac{i}{4}\sum_{k,l} G_{kl}\left(\hat{R}_k\Omega_{lm} + \Omega_{km}\hat{R}_l\right).\end{aligned} \tag{2.27}$$

By exploiting the symmetry of G and antisymmetry of Ω ($\Omega^T = -\Omega$), both terms combine into a single matrix product. Specifically, first term in eq. (2.27) $\sum_{k,l} G_{kl}\Omega_{lm}\hat{R}_k = \sum_{k,l}((G\Omega)^T)_{mk}\hat{R}_k = -\sum_k(\Omega G)_{mk}\hat{R}_k$ (as $\Omega^T = -\Omega$ and $G^T = G$) and similarly for the second term in eq. (2.27). Thus:

$$[\hat{H}_{quad}, \hat{R}] = -\frac{i}{2}\Omega G\hat{R}. \tag{2.28}$$

Now, for the linear term in H, $[\mathbf{b}^T\hat{R}, \hat{R}_m] = \sum_k b_k\frac{i}{2}\Omega_{km} = -\sum_k \frac{i}{2}\Omega_{mk}b_k = -\frac{i}{2}(\Omega\mathbf{b})_m$.

Combining these results and applying the Heisenberg equation $\frac{d\hat{R}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{R}]$ (where constants are absorbed into the definition of Ω), we obtain the linear differential equation:

$$\frac{d\hat{R}}{dt} = \Omega G\hat{R} + \Omega\mathbf{b}. \tag{2.29}$$

Equation 2.29 is a linear differential equation. Defining $A = \Omega G$ and $\mathbf{c} = \Omega\mathbf{b}$, the formal solution is:

$$\hat{R}(t) = S(t)\hat{R}(0) + \mathbf{d}(t), \tag{2.30}$$

where $S(t) = e^{At} = e^{\Omega G t}$ is the homogeneous solution and $\mathbf{d}(t) = \int_0^t e^{\Omega G(t-s)}\Omega\mathbf{b}\,ds$ is the displacement.

Symplectic Nature of $S(t)$: We verify that $S(t)$ is a symplectic transformation, i.e., $S(t)^T \Omega S(t) = \Omega$. Differentiating with respect to time:

$$\frac{d}{dt}\left(S^T \Omega S\right) = S^T \left(A^T \Omega + \Omega A\right) S. \tag{2.31}$$

Substituting $A = \Omega G$ and using $G^T = G, \Omega^T = -\Omega$:

$$\begin{aligned} A^T \Omega + \Omega A &= (\Omega G)^T \Omega + \Omega(\Omega G) \\ &= G^T \Omega^T \Omega + \Omega^2 G \\ &= G(-\Omega)\Omega + \Omega^2 G \\ &= -G\Omega^2 + \Omega^2 G \\ &= -G(-I) + (-I)G \quad (\text{since } \Omega^2 = -I) \\ &= G - G = 0. \end{aligned} \tag{2.32}$$

Since the time derivative is zero, the quantity $S(t)^T \Omega S(t)$ is constant in time. Evaluating at $t = 0$ where $S(0) = I$:

$$S(t)^T \Omega S(t) = S(0)^T \Omega S(0) = I^T \Omega I = \Omega.$$

Thus, $S(t)$ satisfies the definition of a symplectic matrix for all t.

Transformation of Mean and Covariance Matrix: The mean vector is defined as:

$$\bar{R} = \langle \hat{R} \rangle \tag{2.33}$$

From eq. (2.30):

$$\bar{R} \to S\bar{R} + \mathbf{d} \tag{2.34}$$

The covariance matrix is:

$$\sigma_{ij} = \frac{1}{2}\langle \{\hat{R}_i - \bar{R}_i, \hat{R}_j - \bar{R}_j\} \rangle \tag{2.35}$$

Now,

$$\hat{R}' - \bar{R}' = S(\hat{R} - \bar{R}). \tag{2.36}$$